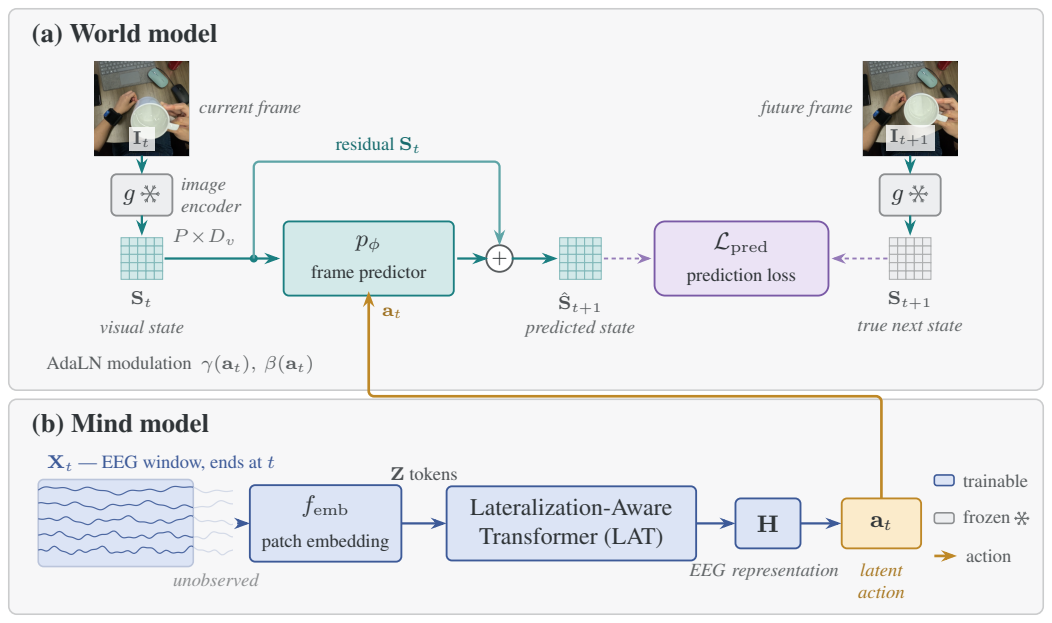

(a) World model
current frame
$\mathbf{I}_t$
residual $\mathbf{S}_t$
g
image encoder
$P \times D_v$
$\mathbf{S}_t$
visual state
p_ϕ
frame predictor
$\mathbf{a}_t$
$\hat{\mathbf{S}}_{t+1}$
predicted state
$\mathcal{L}_{\text{pred}}$
prediction loss
future frame
$\mathbf{I}_{t+1}$
g
$\mathbf{S}_{t+1}$
true next state
AdaLN modulation $\gamma(\mathbf{a}_t)$, $\beta(\mathbf{a}_t)$
(b) Mind model
$\mathbf{X}_t$ — EEG window, ends at t
unobserved
f_{emb}
patch embedding
$\mathbf{Z}$ tokens
Lateralization-Aware Transformer (LAT)
$\mathbf{H}$
EEG representation
$\mathbf{a}_t$
latent action
trainable
frozen
action